Then the half life (in min) is

a) 60 minutes b) 120 minutes

c) 30 minutes d) 15 minutes

Answer the following questions:

1. Define average rate and instantaneous rate.
2. Define rate law and rate constant.
3. Derive integrated rate law for a zero order reaction $A \longrightarrow product$.
4. Define half life of a reaction. Show that for a first order reaction half life is independent of initial concentration.
5. What is an elementary reaction? Give the differences between order and molecularity of a reaction.
6. Explain the rate determining step with an example.
7. Describe the graphical representation of first order reaction.
8. Write the rate law for the following reactions.
   (a) A reaction that is 3/2 order in x and zero order in y.
   (b) A reaction that is second order in NO and first order in Br_2.
9. Explain the effect of catalyst on reaction rate with an example.
10. The rate law for a reaction of A, B and C has been found to be $rate = k[A]^2[B][L]^{3/2}$. How would the rate of reaction change when
    (i) Concentration of [L] is quadrupled
    (ii) Concentration of both [A] and [B] are doubled
    (iii) Concentration of [A] is halved
    (iv) Concentration of [A] is reduced to $\left(\frac{1}{3}\right)$ and concentration of [L] is quadrupled.
11. The rate of formation of a dimer in a second order reaction is 7.5×10^{-3} mol $L^{-1}s^{-1}$ at 0.05 mol L^{-1} monomer concentration. Calculate the rate constant.
12. For a reaction $x + y + z \longrightarrow$ products the rate law is given by $rate = k[x]^{3/2}[y]^{1/2}$ what is the overall order of the reaction and what is the order of the reaction with respect to z.
13. Explain briefly the collision theory of bimolecular reactions.
14. Write Arrhenius equation and explains the terms involved.
15. The decomposition of Cl_2O_7 at 500K in the gas phase to Cl_2 and O_2 is a first order reaction. After 1 minute at 500K, the pressure of Cl_2O_7 falls from 0.08 to 0.04 atm. Calculate the rate constant in s^{-1}.
16. Give two exapmles for zero order reaction
17. Explain pseudo first order reaction with an example.
18. Identify the order for the following reactions

(i) Rusting of Iron

(ii) Radioactive disintegration of $_{92}U^{238}$

(iii) $2A + 3B \longrightarrow$ products ; $rate = k[A]^{1/2}[B]^2$

19. A gas phase reaction has energy of activation 200 kJ mol^{-1}. If the frequency factor of the reaction is 1.6×10^{13} s^{-1}. Calculate the rate constant at 600 K. $\left(e^{-40.09} = 3.8 \times 10^{-18}\right)$

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